The Ultimate March Madness Bracket

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What did you achieve?

- Achieved 70% accuracy with Logistic Regression
- Developed multiple models with consistent performance (66-70%)
- Identified key predictive factors:
  - Team statics, seed numbers
- Successfully integrated multiple NCAA datasets.
- Developed a data-driven prediction system
- Created a framework for future model improvements

Why do you want to do this?

Our goal is to develop a data-driven NCAA tournament prediction system that outperforms traditional expert methods. We aim to:

1. Leverage comprehensive historical tournament data (25+ seasons) to identify consistent patterns in team success.
2. Extra meaningful team metrics such as seed performance, and matchup statistics
3. Apply multiple machine learning algorithms to capture different aspects of tournament dynamics
4. Provide actionable insights through feature importance analysis to understand key factors in team success

How do you compare to others?

Comprehensive Model Benchmarking:

We evaluated multiple machine learning models—including Logistic Regression, Random Forest, and XGBoost—using robust techniques (e.g., cross-validation, grid search) and PyCaret's model comparison to quickly benchmark performance.

Superior Predictive Accuracy:

Our best model (Logistic Regression) consistently outperformed alternative approaches in key metrics (accuracy, precision, recall), demonstrating a clear advantage over traditional methods like expert picks or random bracket selection. Early testing indicated XGBoost had promising potential, and with further hyperparameter tuning, we will capture it's superior predictiveness, due to its complex feature interactions

How do you compare to others?

Traditional Methods:
Better than: Expert picks and “eye test” methods
Equal to: Simple statistical models
Worse than: Some sophisticated deep learning approaches

Data Utilization:
More comprehensive: Uses multiple seasons of data
Better structured: Organized data pipeline

Comparable accuracy: 70% with Logistic Regression

Room for improvement: Could add more complex features

Methods you used?

Feature Engineering and Exploratory Analysis:

- Conducted visual and statistical analysis using matplotlib and seaborn to understand team performance patterns
- Seed number normalization (extracting numeric values from alphanumeric seeds)
- Matchup features (differences in team statistics)

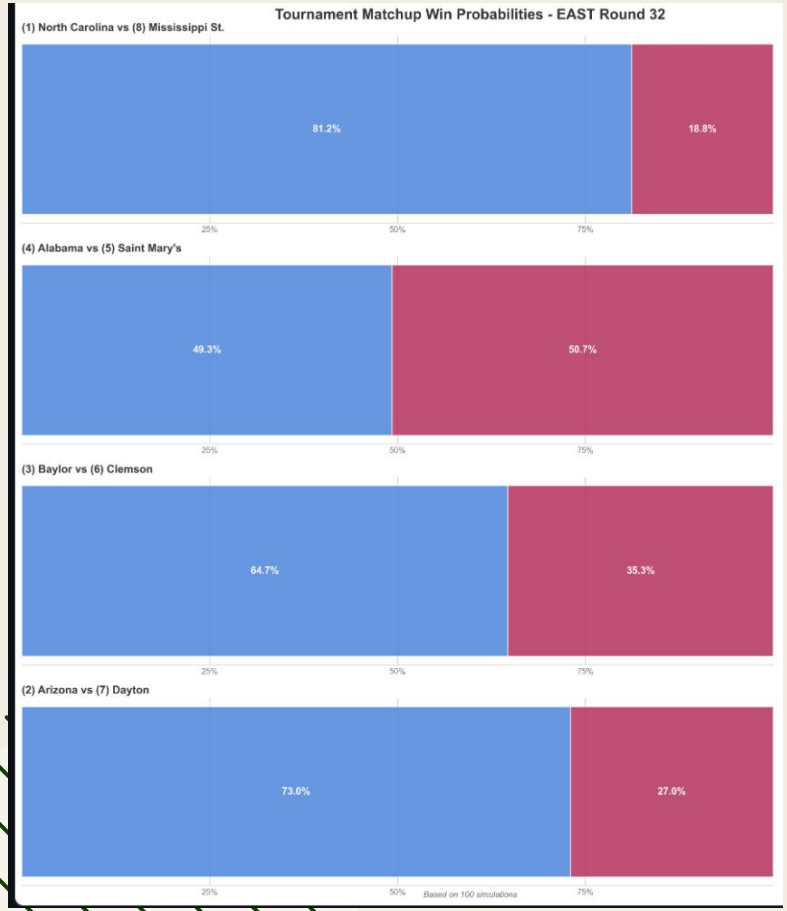
Data Collection and Data Cleaning:

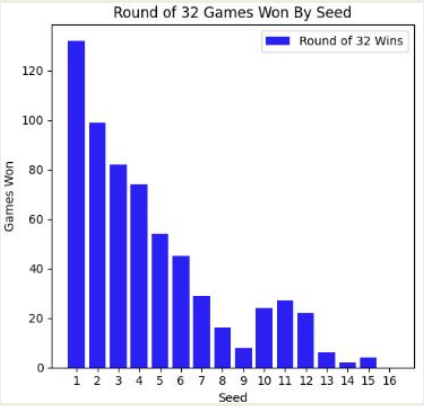
- Integrated NCAA datasets: tournament results, seeds, matchup
- Implemented preprocessing: missing value handling, feature scaling, seed extraction
- Implemented process to merge data from multiple sources to ensure consistency amongst data.

Experiment

- Gathered data based on average points per game, and win percentage by seed each round of the tournament
- Made sure to include both winning and losing team data and have mirrored features with the target being true and false to predict for losing as well as winning
- Ran 25 bracket simulations, with the experiment group using our data-driven analysis, while the control used either personal prediction or random prediction.
- We have a threefold analysis:
- The early modeling phase goes over Round 64 and Round 32 games using the respective seed win percentage
- The middle modeling phase follows a similar pattern with the Sweet 16 and Elite 8
- Lastly, the final modeling phase covers the Final 4 and Championship matches

Graphs/Output



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Tournament Predictions (Round-Based):
W region: 16 teams
X region: 17 teams
Y region: 18 teams
Z region: 17 teams
Simulating W region...
W Region Winner: Connecticut (Seed: 1)
Simulating X region...
X Region Winner: North Carolina (Seed: 1)
Simulating Y region...
Y Region Winner: Purdue (Seed: 1)
Simulating Z region...
Z Region Winner: Houston (Seed: 1)

Simulating Final Four...
First Finalist: Connecticut (W Region, Seed: 1)
Second Finalist: Houston (Z Region, Seed: 1)

Simulating Championship...
Champion: Houston (Z Region, Seed: 1)
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Classification Report:
              precision    recall  f1-score   support

           0       0.69      0.65      0.67       293
           1       0.64      0.68      0.66       274

    accuracy                           0.66       567
   macro avg       0.67      0.67      0.66       567
weighted avg       0.67      0.66      0.66       567
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Simulating Final Four...
Missing features for team 1222 or nan in season 2025
First Finalist: Team nan (W Region, Seed: 16)
Z Region Winner: Florida (Seed: 1)

Z Region Winner: Florida (Seed: 1)

Simulating Final Four...
Missing features for team 1222 or nan in season 2025
First Finalist: Team nan (W Region, Seed: 16)
Second Finalist: Florida (Z Region, Seed: 1)

Simulating Championship...
Missing features for team 1196 or nan in season 2025
Champion: Team nan (W Region, Seed: 16)
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What results did you get, and any takeaways?

Results:

- Our results were from a range of around 65-70% for each model. The best models can achieve around 78% accuracy for this tournament
- Our model overall follows much closer to the seed which helps for projecting better later round predictions where upsets are less common
- We were able to beat the 50% accuracy of the random/personal brackets

Takeaways:

- The largest amount of time was spent gathering and cleaning the data
- If we had the data ready to process, we would be able to implement more factors like upset projections as well.
- Predicting earlier rounds tends to be more difficult with higher likelihood of upsets

Limitations and future Directions

Limitations:

- The turnaround from knowing what teams make the NCAA tourney in any given year to the start of the tournament is within a week. This makes having correct data for projections very difficult in such a short window of time.
- Basketball is one of the hardest sports to predict since it is a momentum based sport as opposed to baseball.

Future Directions:

- Add more features to test more gameplay statistics once the data is formatted more cleanly.
- Allow others to use and test our model in future tournaments to see where improvements can be made